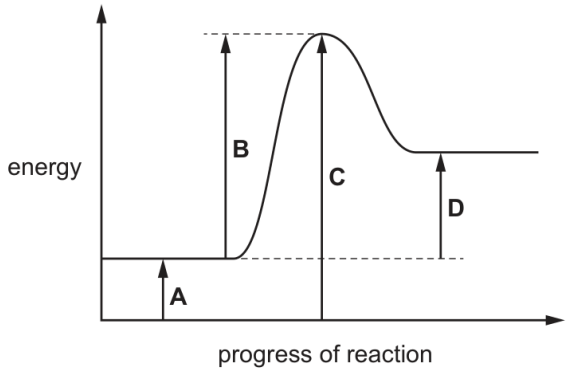
Combined Science: Chemical Energetics

IGCSE Worked Solutions

Each answer below is written using only the concepts taught in this chapter. If a question needs material beyond the chapter, it is flagged for teacher review.

Question 1. *[Combined Science 2025 22]*



Final answer: B

1. Recognise the profile as endothermic: on this energy vs progress-of-reaction graph the products line sits HIGHER than the reactants line, so the reaction takes in energy overall. [from: C5.01 endothermic profile]

2. Recall the chapter definition: activation energy (Ea) is the minimum energy that colliding particles must have for a reaction to occur, shown on the diagram as the height of the 'hump' measured ABOVE the reactants line up to the peak. [from: C5.01 activation energy]

3. Match the arrows to the definition: arrow A shows the reactant energy from the x-axis; arrow C goes from the baseline up to the peak (total energy at the peak, not Ea); arrow D points from the peak down to the products line (energy given out from the activated complex). Only arrow B runs from the reactants dashed line upward to the peak of the hump. [from: C5.01 activation energy]

4. Therefore the label representing the activation energy for the endothermic reaction is B. [from: C5.01 activation energy]

From your chapter: C5.01 activation energy; C5.01 endothermic profile

Question 2. *[Combined Science 2025 12]*

19 Which reaction is **most** endothermic?

	initial temperature / °C	final temperature / °C
A	21	19
B	20	22
C	19	18
D	18	23

Final answer: A — temperature drops by 2 °C, the largest decrease (most endothermic)

1. Use the chapter rule for endothermic reactions: the reaction takes in thermal energy FROM the surroundings, so the surroundings cool down and their temperature DECREASES. That means an endothermic reaction gives a NEGATIVE temperature change ($\Delta T = T_{final} - T_{initial} < 0$). 'Most endothermic' therefore means the LARGEST DROP in temperature (the most negative ΔT), not the largest number. [from: C5.01 endothermic → surroundings cool]

2. Calculate $\Delta T = T_{final} - T_{initial}$ for each option. A: 19 – 21 = –2 °C (cooled, endothermic). B: 22 – 20 = +2 °C (warmed, exothermic — reject). C: 18 – 19 = –1 °C (cooled, endothermic). D: 23 – 18 = +5 °C (warmed, exothermic — reject). Only A and C are endothermic. [from: C5.01 temperature change sign]

3. Compare the two endothermic options by how far the temperature dropped: A fell by 2 °C while C fell by only 1 °C. A larger drop means more thermal energy was taken in from the surroundings, so A is the MOST endothermic. Answer: A. [from: C5.01 comparing endothermic reactions]

From your chapter: C5.01 endothermic → surroundings cool; C5.01 temperature change sign; C5.01 comparing endothermic reactions

Question 3. *[Combined Science 2025 12]*

20 Which statement indicates that a chemical change occurs?

A Energy is absorbed.

B There is a change in colour.

C There is a change of state.

D There is no change in mass.

Final answer: B

1. The question asks which observation uniquely indicates that a CHEMICAL change has occurred, i.e. that new substances have been formed. I will check each option against what the chapter says distinguishes a chemical change from a physical one. [from: C5.01 Exothermic and endothermic reactions]

2. Option A ('Energy is absorbed'): chapter slide 11 point 2 explicitly states that BOTH physical and chemical changes involve changes of energy. For example, melting ice (a physical change) absorbs thermal energy. So 'energy is absorbed' is not a unique sign of a chemical change. Reject A. [from: C5.01 physical-and-chemical changes both involve energy changes]

3. Option C ('There is a change of state'): a change of state (solid to liquid to gas) is a PHYSICAL change because the substance itself is unchanged, only its arrangement of particles changes. No new chemical has been made. Reject C. [from: C5.01 Exothermic and endothermic reactions]

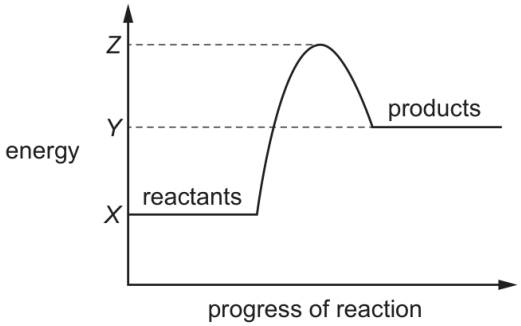
4. Option D ('There is no change in mass'): mass is conserved in essentially every closed-system reaction, physical or chemical, so 'no change in mass' cannot distinguish the two. It is not a positive indicator that anything has happened. Reject D. [from: C5.01 Exothermic and endothermic reactions]

5. Option B ('There is a change in colour'): a new colour means a new substance has been formed - for example copper turning black as it reacts with oxygen to form copper(II) oxide. Forming a new substance is the defining feature of a chemical change, so B is the correct answer. [from: C5.01 Exothermic and endothermic reactions]

From your chapter: C5.01 Exothermic and endothermic reactions; C5.01 physical-and-chemical changes both involve energy changes

Question 4. *[Combined Science 2025 23]*

18 The reaction pathway diagram for a reaction is shown.



Which statement about the reaction is correct?

A The activation energy for this reaction is equal to the value of ($Z - Y$).

B The energy released by this reaction is equal to the value of ($Y - X$).

C The energy used to break bonds is more than the energy released in forming bonds.

D The overall energy change for this reaction is equal to the value of ($Z - X$).

Final answer: C

1. Read the reaction pathway: reactants line sits at level X (bottom), the curve rises to a peak at level Z (top), and the products line sits at level Y, which is above X but below Z. Because the products line is higher than the reactants line, the overall energy change is positive — the reaction is endothermic. [from: Slide 4: exothermic vs endothermic pathway diagrams]

2. Activation energy (Ea) is the energy needed to get from the reactants line up to the peak of the curve. Here that height is (Z – X), not (Z – Y). So option A is wrong, and option D is also wrong because (Z – X) is Ea, not the overall energy change. [from: Slide 6: activation energy = peak – reactants]

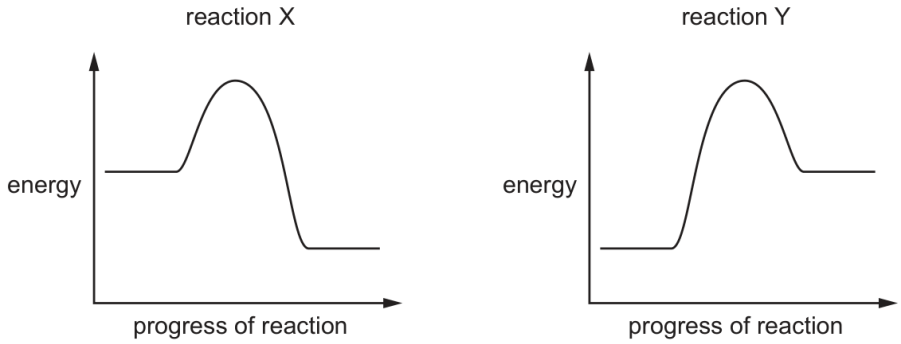
3. Option B says 'energy released = (Y – X)'. Since products are ABOVE reactants, energy is taken in from the surroundings, not released. (Y – X) is the overall energy ABSORBED by this endothermic reaction, so 'released' is the wrong word — B is wrong. [from: Slide 5: endothermic reactions absorb energy]

4. Option C says the energy used to break bonds is more than the energy released in forming bonds. This is exactly the chapter's definition of an endothermic reaction, which is what this diagram shows. So C is correct. [from: Slide 5: endothermic = bond breaking > bond making]

From your chapter: Slide 4: reaction pathway diagrams; Slide 5: endothermic vs exothermic bond-energy rule; Slide 6: activation energy definition

Question 5. *[Combined Science 2025 21]*

20 The reaction pathway diagrams for reaction X and for reaction Y are shown.



Which statement about the reactions is correct?

A Reaction X has a greater activation energy than reaction Y.

B Reaction X is endothermic and reaction Y is exothermic.

C The overall energy change in reaction X is much greater than in reaction Y.

D The temperature increases during reaction X and decreases during reaction Y.

Final answer: D

1. Read the two reaction pathway diagrams. In reaction X, reactants sit HIGHER than products, so the reaction releases energy overall — it is exothermic. In reaction Y, products sit HIGHER than reactants, so the reaction takes in energy overall — it is endothermic. [from: exothermic_endothermic_from_pathway_diagram]

2. Translate 'exothermic' and 'endothermic' into what a student would feel. Exothermic reactions release heat to the surroundings, so the temperature of the surroundings INCREASES during reaction X. Endothermic reactions absorb heat from the surroundings, so the temperature of the surroundings DECREASES during reaction Y. [from: surroundings_temperature_change]

3. Eliminate the other options. A — activation energy is the height of the hump above the reactants; both humps look similar on the diagram, so we cannot claim X has a greater Ea. B — backwards, since X is exothermic and Y is endothermic, not the other way round. C — we have no numerical energy values, and the overall energy gaps look comparable, so 'much greater' is not supported. D — matches the chapter rule 'exothermic warms the surroundings, endothermic cools them', so D is correct. [from: eliminate_distractors_pathway_mcq]

From your chapter: exothermic_endothermic_from_pathway_diagram; surroundings_temperature_change; activation_energy_vs_overall_energy_change

Question 6. *[Combined Science 2025 42]*

5 **(a)** Calcium oxide, CaO, is used to make cement.

Calcium oxide is formed by the decomposition of calcium carbonate, $CaCO_3$, at a very high temperature.

The equation for the reaction is shown.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

(i) Suggest why making cement contributes to global warming.

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..... [2]

(ii) The reaction is endothermic.

Explain what is meant by an endothermic reaction.

Use ideas about making bonds and breaking bonds in your answer.

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..... [3]

Final answer: (a)(i) [2] Making cement thermally decomposes CaCO3, which releases CO2 directly from th

1. a(i) — Identify the carbon-dioxide link. The equation shows CaCO3(s) → CaO(s) + CO2(g): every mole of calcium carbonate that is decomposed releases one mole of CO2 directly into the atmosphere. That is the first source of CO2 from cement manufacture. [from: C5.01 endothermic thermal decomposition of a carbonate (CaCO3 → CaO + CO2)]

2. a(i) — Second source of CO2. Because the decomposition is endothermic, the reactor has to reach a very high temperature (~900 °C). That thermal energy normally comes from burning fossil fuels, which releases additional CO2 into the atmosphere. [from: C5.01 endothermic reactions require a continuous input of thermal energy from the surroundings]

3. a(i) — Link CO2 to global warming. CO2 is a greenhouse gas: it absorbs infrared radiation re-emitted from the Earth's surface, so a higher atmospheric CO2 concentration traps more thermal energy and raises global temperatures. [from: Cross-chapter link (C13 Atmosphere): CO2 as a greenhouse gas — cited here only to justify the C5 endothermic-energy answer]

4. a(ii) — Define endothermic. An endothermic reaction is one that takes IN thermal energy from the surroundings. On an energy pathway diagram the products line sits ABOVE the reactants line and ΔH is positive. The surroundings lose energy, so they feel colder (or, as

here, have to be heated continuously to keep the reaction going). [from: C5.01 endothermic reaction — definition and pathway diagram (products above reactants, $\Delta H > 0$)]

5. a(ii) — Apply the bond-energy rule (MEXOBENDO). Breaking existing bonds in the reactants always ABSORBS energy — bond breaking is ENDOthermic. Making the new bonds in the products always RELEASES energy — bond making is EXOthermic. The overall energy change is: energy absorbed to break bonds – energy released when new bonds form. [from: C5.01 MEXOBENDO — Making = EXO, Breaking = ENDO; bond energies determine ΔH]

6. a(ii) — Put it together for $CaCO_3 \rightarrow CaO + CO_2$. The C–O and Ca–O bonds in $CaCO_3$ have to be broken (energy IN); new bonds are then formed in CaO and in CO_2 (energy OUT). Because the energy needed to break the reactant bonds is GREATER than the energy released on making the product bonds, there is a net intake of thermal energy from the surroundings — that net intake is why the reaction is endothermic and needs a very high temperature to proceed. [from: C5.01 endothermic condition — energy absorbed (breaking) > energy released (making) $\Rightarrow \Delta H$ positive]

From your chapter: C5.01 endothermic vs exothermic reactions; C5.01 MEXOBENDO (Making = EXO, Breaking = ENDO); C5.01 energy change = energy absorbed (breaking) – energy released (making); C5.01 thermal decomposition of $CaCO_3$ as a high-temperature endothermic reaction; Cross-chapter: C13 Atmosphere — CO_2 as a greenhouse gas

Question 7. *[Combined Science 2025 42]*

(iii) Fig. 5.1 shows the incomplete reaction pathway diagram for an endothermic reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, E_a
- a labelled arrow to show the overall energy change of the reaction.

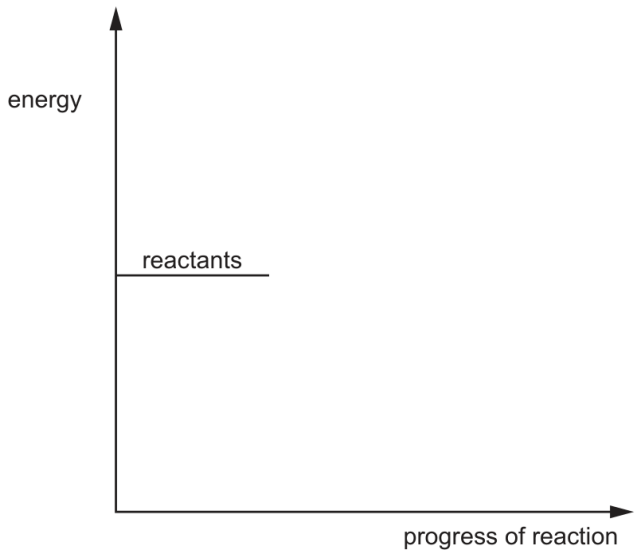


Fig. 5.1

[3]

(b) Calcium oxide reacts with water to form calcium hydroxide.

(i) The ions in calcium hydroxide are Ca^{2+} and OH^-.

Deduce the formula for calcium hydroxide.

..... [1]

(ii) Aqueous calcium hydroxide is an alkali.

State the colour of methyl orange indicator in aqueous calcium hydroxide.

..... [1]

[Total: 10]

Final answer: (a)(iii) [3] Complete the endothermic pathway on Fig. 5.1: draw a horizontal PRODUCTS line

1. a(iii) — Position the products line. The reaction is endothermic, so the products must sit at a HIGHER energy than the reactants. On Fig. 5.1 draw a horizontal line labelled 'products' ABOVE the existing reactants line. The existing hump connecting them has its peak above even the products line — the activated complex. [from: C5.01 endothermic reaction pathway — products above reactants, ΔH positive]

2. a(iii) — Label the activation energy. Draw a vertical (upward) arrow that STARTS on the reactants line and ENDS at the very top of the hump (the peak), and label it Ea (activation energy). Ea is the minimum energy colliding reactant particles must gain for the reaction to occur — it is the height of the hump above the REACTANTS line, not above the products line. [from: C5.01 activation energy Ea — height from reactants line up to the peak]

3. a(iii) — Label the overall energy change. Draw a vertical (upward) arrow from the reactants line to the products line and label it 'overall energy change' (accept ΔH, 'energy taken in', 'enthalpy change'). Because the reaction is endothermic the arrow points UP and ΔH is positive — this is the net thermal energy absorbed from the surroundings. [from: C5.01 overall energy change ΔH — reactants line to products line; endothermic ⇒ upward arrow, ΔH positive]

4. b(i) — OUT OF C5 SCOPE (belongs to C2 Bonding / C8 Acids, Bases & Salts). Using the charges to balance: Ca^{2}■ has a 2+ charge; OH■ has a 1– charge. To make a neutral compound you need two OH■ for every one Ca^{2}■, giving Ca(OH)2. Brackets are needed because the OH stays together as a single ion. [from: OUT OF C5 SCOPE — see C8 Acids, Bases & Salts (/ C2 Bonding) for deducing ionic formulae]

5. b(ii) — OUT OF C5 SCOPE (belongs to C8 Acids, Bases & Salts). Aqueous calcium hydroxide (limewater) is an alkali (pH ~12). Methyl orange is YELLOW in alkaline solutions (red in acid, yellow in alkali; the mid-point is orange). So the colour is yellow. [from: OUT OF C5 SCOPE — see C8 Acids, Bases & Salts for indicator colours (methyl orange, litmus, phenolphthalein)]

From your chapter: C5.01 endothermic reaction pathway diagram (products above reactants); C5.01 activation energy Ea (reactants line to peak); C5.01 overall energy change ΔH (reactants line to products line; endothermic ⇒ upward, ΔH positive); OUT OF C5 SCOPE: C8 Acids, Bases & Salts — ionic formulae and indicator colours

Question 8. *[Combined Science 2025 22]*

17 Which statement explains why a reaction is exothermic?

A More thermal energy is absorbed by breaking bonds than is released by making bonds.

B More thermal energy is absorbed by making bonds than is released by breaking bonds.

C More thermal energy is released by breaking bonds than is absorbed by making bonds.

D More thermal energy is released by making bonds than is absorbed by breaking bonds.

Final answer: D — more thermal energy released by making bonds than absorbed by breaking bonds

1. Recall the MEXOBENDO rule: Making bonds is EXOthermic (releases thermal energy), Breaking bonds is ENDOthermic (absorbs thermal energy). So 'absorbed' pairs with breaking, and 'released' pairs with making. [from: MEXOBENDO — bond breaking endothermic / bond making exothermic]

2. The net energy change of a reaction = (energy released when new bonds are MADE) – (energy absorbed when old bonds are BROKEN). This is just the sum of endothermic and exothermic bond-change steps. [from: C5.01 net energy change = energy released (making) – energy absorbed (breaking)]

3. A reaction is overall EXOthermic when it gives out more energy than it takes in, i.e. energy released by making new bonds > energy absorbed by breaking old bonds. That matches option D. Option B inverts MEXOBENDO (bond making does not absorb). Options A and C also swap the 'absorbed/released' pairings, so they violate MEXOBENDO. [from: C5.01 exothermic condition: released-by-making > absorbed-by-breaking]

From your chapter: MEXOBENDO — bond breaking endothermic / bond making exothermic; C5.01 net energy change = energy released (making) – energy absorbed (breaking); C5.01 exothermic condition: released-by-making > absorbed-by-breaking

Question 9. *[Combined Science 2025 22]*

18 Which statement explains why the rate of a reaction is greater at a higher temperature?

A The activation energy is higher.

B More colliding particles have the minimum energy to react.

C The frequency of collisions between reacting particles decreases.

D There are more reacting particles per unit volume.

Final answer: B

1. From the chapter's KEY WORDS, activation energy (Ea) is 'the minimum energy that colliding particles must have to react successfully'. The reaction rate rises with temperature only when more of the colliding particles can meet this Ea threshold. I will test each option against that definition. [from: C5.01 KEY WORDS activation energy definition]

2. Option A ('The activation energy is higher'): Ea is a fixed property of the specific reaction - it depends on the bonds that must be broken to start the reaction. Heating the mixture does NOT change Ea itself; it only changes how many particles have enough energy to reach Ea. Reject A. [from: C5.01 activation energy]

3. Option C ('The frequency of collisions ... decreases'): higher temperature means particles move faster (more kinetic energy), so they collide MORE often per second, not less. This option has the direction reversed. Reject C. [from: C5.01 activation energy]

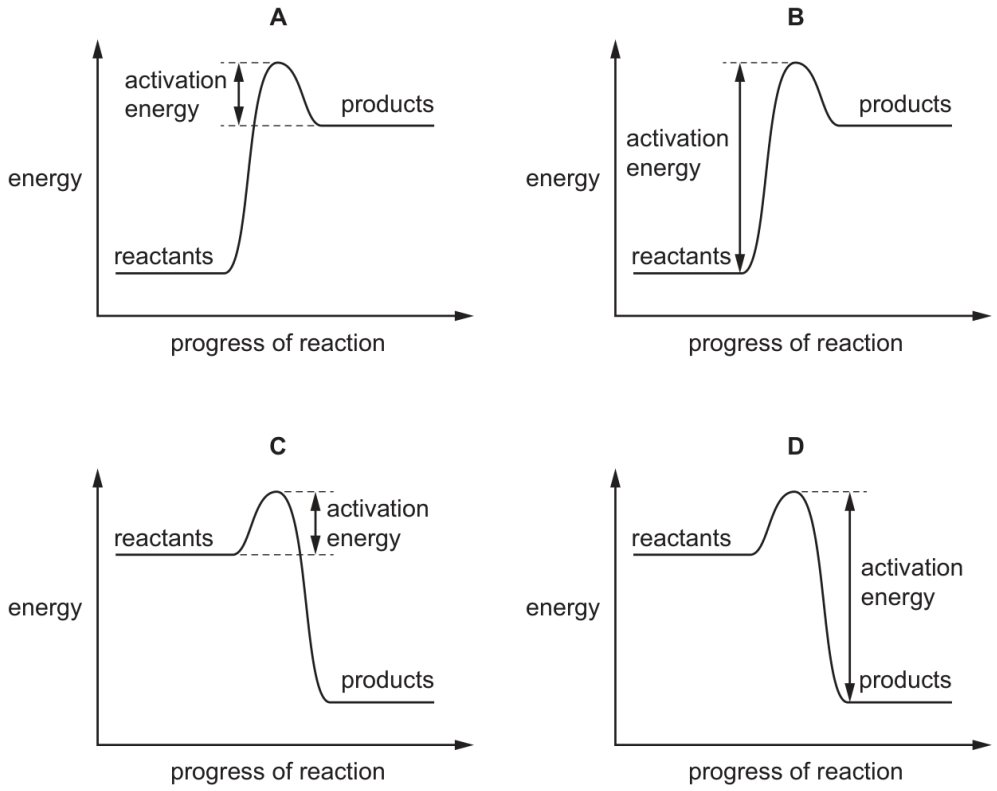
4. Option D ('There are more reacting particles per unit volume'): changing temperature does not change how many particles are in the container - that would be a change in concentration, not temperature. So D describes the wrong variable. Reject D. [from: C5.01 activation energy]

5. Option B ('More colliding particles have the minimum energy to react'): at higher temperature the average kinetic energy of the particles rises, so a greater fraction of colliding particles now have energy greater than or equal to Ea. Those extra successful collisions are exactly why the rate goes up. This matches the chapter definition of activation energy. Choose B. [from: C5.01 KEY WORDS activation energy definition]

From your chapter: C5.01 activation energy; C5.01 KEY WORDS activation energy definition

Question 10. *[Combined Science 2025 22]*

18 Which reaction pathway diagram shows the activation energy for an exothermic reaction?



Final answer: C — diagram C is the only one that both shows an exothermic profile (products drawn lower

1. Recall what activation energy (Ea) is: the minimum energy that colliding particles must have to react. On a reaction pathway diagram it is drawn as the vertical height from the reactants line UP to the top of the hump (the energy barrier), no matter whether the reaction is exothermic or endothermic. [from: C5.01 activation energy]

2. Recall what an exothermic profile looks like: exothermic reactions release thermal energy to the surroundings, so the products line is drawn LOWER than the reactants line (ΔH is negative). Diagrams A and B show products higher than reactants, so they are endothermic and can be eliminated. [from: C5.01 exothermic reaction profile]

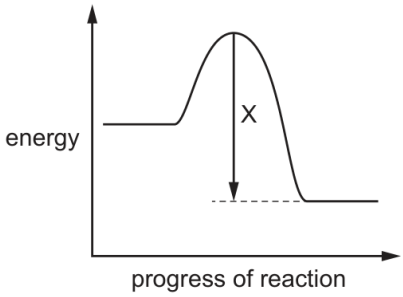
3. That leaves C and D, which both show products lower than reactants (exothermic). Check which arrow is the activation energy arrow: it must go from the reactants horizontal line UP to the peak of the hump (bond breaking is endothermic and takes the system uphill to the peak — MEXOBENDO). [from: C5.01 activation energy]

4. In diagram D the labelled arrow goes from the peak DOWN to the products line — that is the total energy released ($|\Delta H|$), not the activation energy. In diagram C the labelled arrow goes from the reactants line UP to the top of the hump, which is exactly the activation energy. So C satisfies both conditions: exothermic profile AND Ea drawn correctly. [from: C5.01 exothermic reaction profile]

From your chapter: C5.01 activation energy; C5.01 exothermic reaction profile

Question 11. *[Combined Science 2025 21]*

15 A reaction pathway diagram for a reaction is shown.



Which statement describes and explains energy change X?

A Thermal energy is given out as bonds break.

B Thermal energy is given out as bonds form.

C Thermal energy is taken in as bonds break.

D Thermal energy is taken in as bonds form.

Final answer: B

1. Read the diagram: on a reaction pathway diagram, energy is on the y-axis and progress of reaction is on the x-axis. The curve rises from the reactants line to a peak (the activated complex / transition state) and then falls to the products line. The arrow labelled X points DOWN from the peak to the products line, so X is the energy released during the second half of the reaction. [from: C5.01 reaction pathway diagrams]

2. Identify what is happening at the peak: at the top of the hump the original reactant bonds have been fully broken and the atoms are momentarily free (the transition state). Bond BREAKING is what took energy IN on the way UP from reactants to the peak (the left half, not X). [from: C5.01 bond breaking endothermic]

3. Identify what is happening on the descent (X): as the reaction proceeds past the peak, the free atoms rearrange and FORM the new product bonds. Making new bonds RELEASES thermal energy to the surroundings, so the curve drops. That is exactly what X represents: energy given out as bonds form (exothermic step of the reaction). [from: C5.01 bond making exothermic]

4. Match to the options using the mnemonic MEXOBENDO (Making bonds = EXOthermic, Breaking bonds = ENDOthermic). X is on the descent, so energy is given out, and this happens because bonds are forming. Option A (given out, bonds break) and C (taken in, bonds break) wrongly say bonds are breaking; D (taken in, bonds form) flips the energy direction. Only B matches both pieces: thermal energy is given out AS bonds form. [from: MEXOBENDO]

From your chapter: C5.01 reaction pathway diagrams; C5.01 bond making exothermic; C5.01 bond breaking endothermic; MEXOBENDO

Question 12. *[Combined Science 2025 41]*

(c) Fig. 4.2 is a reaction pathway diagram for a reaction catalysed by copper.

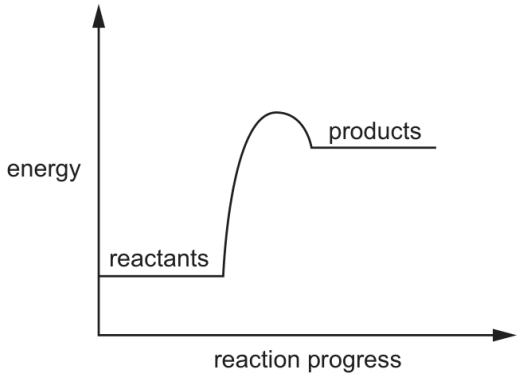


Fig. 4.2

(i) Identify the type of chemical reaction shown in Fig. 4.2.

Explain your answer.

type of chemical reaction .....

explanation .....

.....

..... [2]

(ii) Draw an arrow on Fig. 4.2 to show the activation energy for this reaction. [1]

(iii) Define activation energy.

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..... [2]

(iv) Describe what is meant by a catalyst.

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..... [2]

[Total: 11]

Final answer: c(i) Endothermic — products are at a higher energy level than the reactants, so energy is ab

1. c(i), mark 1: Read the reaction pathway diagram in Fig. 4.2. The reactants (Cu + O2) sit on a lower energy level than the product (CuO) — i.e. the right-hand side of the diagram is higher than the left. Because the products end up at a higher energy than the reactants, the reaction has taken energy in from the surroundings. The name for this is ENDOTHERMIC.

[from: Chemical Energetics — endothermic reactions have products higher than reactants on a reaction pathway

diagram (energy absorbed, ΔH positive).]

2. c(i), mark 2: Justify using the diagram. Say: 'products are at a higher energy level than the reactants, so overall energy is absorbed from the surroundings.' Do NOT say exothermic — that is the opposite and would lose both marks. [from: Chemical Energetics — exo vs endo is decided by comparing reactant and product energy levels on the diagram.]

3. c(ii), mark 1: The activation energy (Ea) is the energy barrier between reactants and the peak of the hump. On Fig. 4.2 you draw a single vertical arrow that STARTS on the reactants' horizontal energy line (left flat line) and ENDS at the top of the hump. A common slip is to draw the arrow from the product line, or to make it the full reactant-to-product difference — that is ΔH, not Ea. Label it 'Ea' or 'activation energy' if space allows. [from: Chemical Energetics — activation energy is represented on a reaction pathway diagram by the height from the reactant energy level up to the top of the hump.]

4. c(iii), mark 1: Start the definition with 'minimum'. 'The minimum amount of energy…'. The word 'minimum' (or an equivalent like 'least') is essentially required — saying 'the energy needed' without 'minimum' is often marked down. [from: Chemical Energetics — activation energy is the MINIMUM energy required.]

5. c(iii), mark 2: Finish the definition by saying whose energy it is and what it does. Full definition: 'the minimum energy that colliding reactant particles must have in order for a reaction to occur (or: to start breaking bonds).' Acceptable alternate phrasings: 'the minimum energy particles need on collision to react' or 'the minimum energy needed to start a chemical reaction.' [from: Chemical Energetics — activation energy is the minimum energy for colliding particles to react (to begin bond breaking).]

6. c(iv), mark 1: First mark is for the effect: 'a catalyst increases (speeds up) the rate of a chemical reaction.' Accept: 'makes the reaction go faster'. Do NOT just say 'helps the reaction' — that's too vague. [from: Chemical Energetics (borderline with Rates) — the diagram in Fig. 4.2 is for a catalysed reaction; a catalyst speeds up a reaction.]

7. c(iv), mark 2: Second mark is for the 'not consumed' idea: 'and is not used up / is chemically unchanged at the end of the reaction.' Accept: 'can be recovered at the end', 'its mass is the same before and after'. A stronger (but not required) version adds 'by providing an alternative pathway with a lower activation energy' — this would also score. [from: Chemical Energetics — reaction pathway diagrams can show a second, lower hump for a catalysed route; the catalyst itself is unchanged overall.]

From your chapter: Chemical Energetics: reaction pathway diagrams show reactant energy level, product energy level, and the activation-energy hump between them.; Chemical Energetics: endothermic = products higher than reactants (energy absorbed); exothermic = products lower than reactants (energy released).; Chemical Energetics: activation energy = the minimum energy reactant particles need (on collision) for the reaction to occur.; Chemical Energetics (catalyst, lightly covered): a catalyst speeds up a reaction and is not used up; in a pathway diagram it lowers the activation-energy hump but leaves ΔH unchanged.

Question 13. *[Combined Science 2025 31]*

6 **(a)** Fig. 6.1 shows the combustion of magnesium ribbon in air.

A white powder of magnesium oxide is formed.

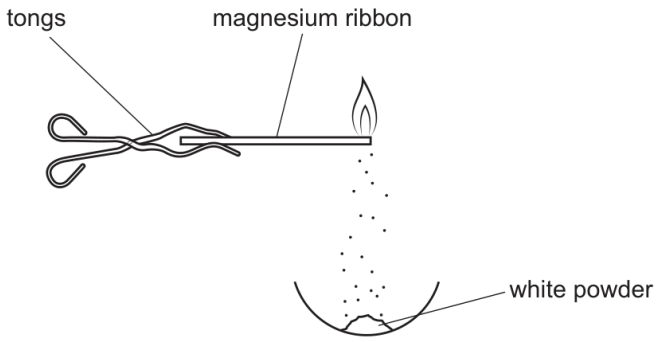


Fig. 6.1

(i) Name the gas that reacts with the magnesium.

..... [1]

(ii) Magnesium oxide is a metallic oxide.

Circle the term that describes this type of oxide.

acidic **aqueous** **basic** **carbonate** **solvent**

[1]

(iii) This reaction is exothermic.

Describe what is meant by exothermic.

.....

..... [1]

Final answer: a(i) oxygen; a(ii) basic; a(iii) exothermic means (thermal) energy/heat is released (given out)

1. a(i) The burning magnesium ribbon reacts with a gas in the air. Air is ~21% oxygen, and the product is magnesium oxide (MgO) — a metal-oxygen compound — so the reacting gas must be oxygen. Worked Example C5.01 writes this combustion as Mg + O2 -> MgO, confirming oxygen is the reactant. Answer: oxygen. [from: C5.01 Worked Example: combustion of Mg uses O2 as reactant gas]

2. a(ii) Magnesium is a metal, and metallic oxides are basic (they react with acids to form a salt + water; e.g. MgO + 2HCl -> MgCl2 + H2O). The other terms don't fit: 'acidic' is for non-metal oxides (CO2, SO2), 'aqueous' means dissolved in water, 'carbonate' contains CO3^2-, and 'solvent' is a dissolving medium. Answer: circle 'basic'. (Note: this sub-part is general acid-base chemistry, not chemical energetics.) [from: out-of-scope for C5 Chemical Energetics (acid/base classification)]

3. a(iii) 'Exothermic' describes a reaction that transfers (thermal) energy out to the surroundings. During Mg combustion you can see and feel energy leaving the system — a bright white light and heat are given out, so the surroundings warm up (temperature of the surroundings rises). Answer: exothermic means heat (thermal energy) is released/given out

to the surroundings, causing the surroundings to get hotter. [from: C5.01 Worked Example: Mg + O2 releases heat and light => exothermic; surroundings warm up]

From your chapter: C5.01 Worked Example C5.01 — combustion of magnesium (2Mg + O2 -> 2MgO) releases heat and light; exothermic reactions transfer energy to the surroundings so the surroundings warm up; C5 definition: exothermic = energy given out to surroundings

Question 14. *[Combined Science 2025 42]*

(iii) Tick (✓) **all** the statements that are true for aqueous sodium hydroxide.

It is a covalent substance. ☐

It turns universal indicator purple. ☐

It reacts with dilute sulfuric acid to form sodium sulfate. ☐

It is a base but **not** an alkali. ☐

[2]

(b) The reaction between aqueous sodium hydroxide and dilute hydrochloric acid transfers thermal energy to the surroundings.

(i) Give **one** conclusion about the energy involved in bond breaking and bond making in this reaction.

.....

..... [1]

(ii) Complete Fig. 4.2 to show the reaction pathway diagram for this reaction.

Label:

- the reactants and products
- the overall energy change using an arrow.

Fig. 4.2

[3]

[Total: 10]

Final answer: a(iii) tick statements 2 and 3 only; b(i) more energy is released when bonds are made (in pro

1. a(iii) — check each statement for aqueous NaOH. (1) 'It is a covalent substance' — FALSE: NaOH is ionic (Na+ ions held to OH- ions by ionic bonding). (2) 'It turns universal indicator purple' — TRUE: aqueous NaOH is a strong alkali, pH ~13–14, so universal indicator goes dark blue / purple. (3) 'It reacts with dilute sulfuric acid to form sodium sulfate' — TRUE: $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$ (classic acid + alkali → salt + water). (4) 'It is a base but not an alkali' — FALSE: NaOH dissolves in water, and a soluble base IS an alkali. So tick statements 2 and 3 only (2 marks: 1 per correct tick; a wrong tick cancels a correct tick in most CIE schemes). [from: C5.01 properties of alkalis]

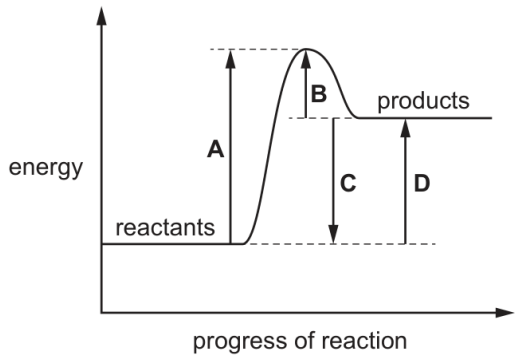
2. b(i) — the stem tells us the reaction 'transfers thermal energy to the surroundings', so it is EXOTHERMIC (the surroundings warm up). Use MEXOBENDO (slide 6): bond MAKING is EXOthermic (gives energy out), bond BREAKING is ENDOthermic (takes energy in). For the reaction to release energy overall, the energy GIVEN OUT when new bonds form in the products must be GREATER than the energy TAKEN IN to break the bonds in the reactants. One-line conclusion: 'More energy is released when bonds are made than is absorbed when bonds are broken.' [from: MEXOBENDO]

3. b(ii) — complete the reaction pathway diagram on Fig. 4.2. Because the reaction is exothermic, the products must end LOWER in energy than the reactants. (a) Draw a short horizontal line on the left at a higher energy — label it 'reactants: NaOH + HCl'. (b) Draw a short horizontal line on the right at a LOWER energy — label it 'products: NaCl + H2O'. (Connect them with a smooth curve if you like, but the two labelled levels are what earn the mark.) (c) Between the two levels draw a vertical arrow pointing DOWNWARDS from the reactants level to the products level, and label it 'overall energy change' (or ΔH, or 'energy given out'). The arrow points down because energy is released to the surroundings. Mark scheme: 1m reactants above products (exothermic shape); 1m reactants and products correctly labelled; 1m arrow showing overall energy change, labelled and pointing downwards. [from: C5.01 exothermic profile]

From your chapter: C5.01 exothermic profile; C5.01 bond making exothermic; MEXOBENDO; C5.01 properties of alkalis

Question 15. *[Combined Science 2025 23]*

19 Which arrow on the reaction pathway diagram represents the overall energy change for the reaction?



Final answer: D — the arrow that goes from the reactants horizontal line directly to the products horizontal

1. The overall energy change of a reaction is the difference between the energy of the products and the energy of the reactants. On a reaction pathway diagram this is the vertical gap from the reactants line to the products line. [from: C5.01 enthalpy change]

2. It is NOT the activation energy (which goes from the reactants line UP to the peak), and it is NOT the energy released from the peak down to the products line. [from: C5.01 activation energy]

3. Looking at the four labelled arrows, the one that connects the reactants horizontal line straight to the products horizontal line — without touching the peak — is arrow D. So D is the overall energy change. [from: C5.01 reaction pathway diagram]

From your chapter: C5.01 enthalpy change; C5.01 activation energy; C5.01 reaction pathway diagram

Question 16. *[Combined Science 2025 42]*

(c) The reaction between aluminium and iron(III) oxide is exothermic.

A flame is used to provide the activation energy to start the reaction.

Fig. 5.1 shows the incomplete reaction pathway diagram for this reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, E_a
- a labelled arrow to show the overall energy change of the reaction.

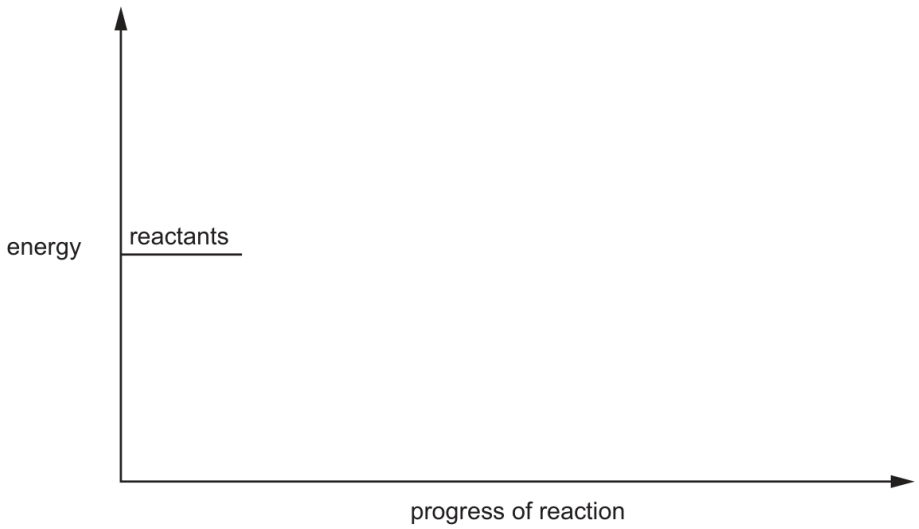


Fig. 5.1

[3]

[Total: 9]

Final answer: Completed exothermic reaction pathway on Fig. 5.1: (i) draw a horizontal products line BELO

1. Complete the pathway. The reaction is exothermic, so the products must be at a lower energy than the reactants. On Fig. 5.1, draw a horizontal line labelled 'products' BELOW the existing reactants line. Join the reactants line to the products line with a smooth hump whose

peak is clearly ABOVE the reactants line (this peak is the transition state that the flame supplies energy to reach). [from: C5.01 reaction pathway diagram – exothermic profile (products lower than reactants, activation-energy hump)]

2. Label the activation energy. Draw a vertical arrow from the reactants line up to the top of the hump (the peak) and label it 'Ea' (or 'activation energy'). This represents the minimum energy the reactant particles must gain – supplied here by the flame – before the reaction can proceed. [from: C5.01 activation energy – Ea is the height from the reactants line to the peak]

3. Label the overall energy change. Draw a vertical arrow from the reactants line DOWN to the products line and label it 'overall energy change' (accept 'energy released', 'ΔH', 'enthalpy change', and note that ΔH is negative because the reaction is exothermic). This is the net energy released to the surroundings as the $Al + Fe_2O_3 \rightarrow Al_2O_3 + Fe$ reaction proceeds. [from: C5.01 enthalpy change ΔH – exothermic reactions release energy, ΔH is negative]

From your chapter: C5.01 reaction pathway diagram (exothermic profile); C5.01 activation energy Ea; C5.01 overall energy change ΔH (exothermic = negative); Worked Example C5.01 ($Mg + O_2 \rightarrow MgO$) – template exothermic pathway with flame-supplied Ea

Question 17. *[Combined Science 2025 12]*

16 The initial and the final temperatures of four different reactions are recorded.

Which reaction is the **most** exothermic?

	initial temperature / °C	final temperature / °C
A	19	16
B	20	19
C	22	24
D	24	25

Final answer: C — temperature increases by +2 °C (largest exothermic temperature rise)

1. An exothermic reaction transfers thermal energy FROM the reacting chemicals TO the surroundings (the water/solution in the beaker). So the surroundings warm up and the measured temperature RISES. In symbols, ΔT = T(final) – T(initial) is POSITIVE for an exothermic reaction. An endothermic reaction does the opposite: it takes in thermal energy from the surroundings, so the surroundings cool down and ΔT is negative. [from: C5.01 exothermic warms surroundings (ΔT positive); endothermic cools surroundings (ΔT negative)]

2. Work out ΔT = T(final) – T(initial) for each row and tag it. A: 16 – 19 = –3 °C $\rightarrow$ temperature FELL, so A is endothermic (reject). B: 19 – 20 = –1 °C $\rightarrow$ temperature FELL, so B is endothermic (reject). C: 24 – 22 = +2 °C $\rightarrow$ temperature ROSE, so C is exothermic. D: 25 – 24 = +1 °C $\rightarrow$ temperature ROSE, so D is exothermic. [from: C5.01 compute ΔT and classify exo vs endo by sign]

3. 'Most exothermic' means the reaction that releases the most thermal energy to the surroundings, which shows up as the LARGEST temperature rise. Comparing only the exothermic options (C and D): C gives ΔT = +2 °C and D gives ΔT = +1 °C. +2 > +1, so C is the most exothermic. Common trap: option A has the biggest change in size (|–3| = 3), but that change is a DROP, so A is the most endothermic, not the most exothermic. [from: C5.02 larger positive ΔT = more energy released = more exothermic]

From your chapter: C5.01 exothermic reaction warms surroundings (ΔT positive); C5.01 endothermic reaction cools surroundings (ΔT negative); C5.02 magnitude of temperature change indicates amount of energy transferred; C5.02 compare positive ΔT values to judge which reaction is most exothermic

Question 18. *[Combined Science 2025 13]*

16 The initial and the final temperatures of four different reactions are recorded.

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3. 'Most exothermic' means the reaction that releases the most thermal energy to the surroundings, which shows up as the LARGEST temperature rise. Comparing only the exothermic options (C and D): C gives ΔT = +2 °C and D gives ΔT = +1 °C. +2 > +1, so C is the most exothermic. Common trap: option A has the biggest change in size (|–3| = 3), but that change is a DROP, so A is the most endothermic, not the most exothermic. [from: C5.02 larger positive ΔT = more energy released = more exothermic]

From your chapter: C5.01 exothermic reaction warms surroundings (ΔT positive); C5.01 endothermic reaction cools surroundings (ΔT negative); C5.02 magnitude of temperature change indicates amount of energy transferred; C5.02 compare positive ΔT values to judge which reaction is most exothermic

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